

JUNE 2024

Investing in Science and Technology

The United States Needs to Up Its Game

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Sujai Shivakumar

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A Report of the CSIS Renewing American Innovation

CSIS

CENTER FOR STRATEGIC &
INTERNATIONAL STUDIES

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Acknowledgments

This report is made possible through general support to CSIS. No direct sponsorship contributed to this report.

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Introduction

The United States is facing a challenge to its global leadership in science and technology that is more serious than any it has confronted since gaining that position after World War II.

Within the relatively short span of two decades, China has emerged as a formidable rival, mounting a concerted drive to dominate key technology-intensive sectors and increasingly matching or exceeding the United States in resources committed. At the same time, the U.S. Congress is bogged down in protracted struggles over public spending and the role of government that have engulfed needed investments in science, research, development, and education—the foundation of U.S. economic strength.

The U.S. private sector remains more innovative than its Chinese competitors, but its efforts are focused on developing consumer-oriented products. Meanwhile, the centralized Chinese system concentrates sustained, long-term government support in technology areas that have direct security-related implications—namely, artificial intelligence (AI), quantum information science, and semiconductors.

The U.S.-China technology rivalry, with its inherent asymmetries, has manifest strategic implications that can ultimately determine the outcome of a potential military confrontation, should one come about. Moreover, as U.S. technology leadership declines, the risks of economic and military challenges will rise.

The limited appreciation of the nature of this rivalry with China was recently underscored by the failure of the U.S. Congress to appropriate the funds previously authorized for the largest part of the CHIPS and Science Act of 2022, while simultaneously cutting the budgets of key federal science

and technology agencies for fiscal year 2024. This is all the more disappointing, not to say alarming, given that the same week, China announced a 10 percent increase in its already significant levels of public spending on research and development (R&D).¹

Foundational Funding for CHIPS and Science on Hold

In a major shift in U.S. policy, in August 2022 Congress passed the CHIPS and Science Act (“CHIPS Act”), which authorized and appropriated nearly \$52 billion in federal investments to promote the creation and expansion of semiconductor manufacturing capacity and microelectronics R&D on U.S. soil. The act also authorized tax credits initially valued (pre-passage) by the Congressional Budget Office at \$46 billion through 2031 for investments in domestic semiconductor manufacturing facilities.²

Welcoming this legislation, the White House said it “will boost American semiconductor research, development and production,” ensuring U.S. leadership in this foundational technology.³

In a concrete manifestation of the program, in March 2024 the Commerce Department announced \$8.5 billion in federal grants for U.S. chipmaker Intel Corporation, to be augmented by \$11 billion in concessional loans to support the company’s creation of new chipmaking capacity in the United States.⁴ That came one month after the administration announced a \$1.5 billion grant for foundational chips to GlobalFoundries.⁵ Since March, the process has accelerated, with awards of \$6.4 billion to Samsung, \$6.6 billion to Taiwan Semiconductor Manufacturing Company (TSMC), and \$6.14 billion to Micron Technology for advanced DRAM (memory chip) production in the United States.⁶ These grants have unquestionably contributed to a massive surge in U.S.-based investments in semiconductor production.

In addition to the \$39 billion to support chip manufacturing, the CHIPS Act authorized a far larger expenditure of \$174 billion for investment in the science, engineering, R&D, education, and workforce development that will underpin future advances in chip design and manufacturing,

with the goal of assuring future U.S. leadership in this critical enabling industry. The principal federal recipients of these funds are to be the National Science Foundation (NSF), the Department of Energy (DOE), the Department of Commerce's National Institute of Standards and Technology (NIST), and the Economic Development Administration (EDA).

In contrast to the funds allocated to chip manufacturing, however, these funds were authorized but not *appropriated* by Congress.

The FY 2024 congressional appropriation did not deliver on the CHIPS Act authorizations for science.⁷ Under the CHIPS Act, the NSF was to receive \$81 billion over five years, doubling its budget by FY 2027, starting with \$15.6 billion in FY 2024. Instead, in the FY 2024 budget, the NSF will receive just \$9 billion, 42 percent of the CHIPS Act target, compounded by an 8 percent cut in its current budget.⁸ Commenting on these reductions, an NSF spokesperson observed that “it is difficult to place this in the context of the rapid, large-scale science investments by our competitors such as China with the express purpose of outcompeting the United States.”⁹

The FY 2024 budget also features additional funding cuts funding to NIST and the National Aeronautics and Space Administration (NASA). And looking forward, President Joe Biden's proposed budget for FY 2025 calls for programmatic increases for NSF and NIST, but not at the levels envisioned by the CHIPS Act, in a retreat from the goals set out in the act.¹⁰

There is very little in the CHIPS Act benefiting the U.S. semiconductor design industry, which has recently lost major markets in China as a result of tightened U.S. export controls on chip technology to that country. U.S. design firms are not eligible for direct federal funding or investment tax credits under the act, yet they are a major source of competitive strength for the United States.

Sadly, the failure to follow through on the promises of the CHIPS Act is not new. Much of the America COMPETES Act of 2007, which also authorized funding for the NSF and other federal science agencies, was never implemented.

Amid the current climate of geopolitical competition with China, a similar failure to fund the CHIPS Act provisions would represent a serious miscalculation.¹¹ In the words of Matt Hourihan, a science specialist at the Federation of American Scientists, cuts “to agencies like NSF and NIST are frankly unconscionable in an era when we should be enhancing support for U.S. scientists and engineers.”¹²

China's Innovation Drive

At a time when science and innovation underpin both competitiveness and national strength, the comparisons with Chinese efforts are disturbing.

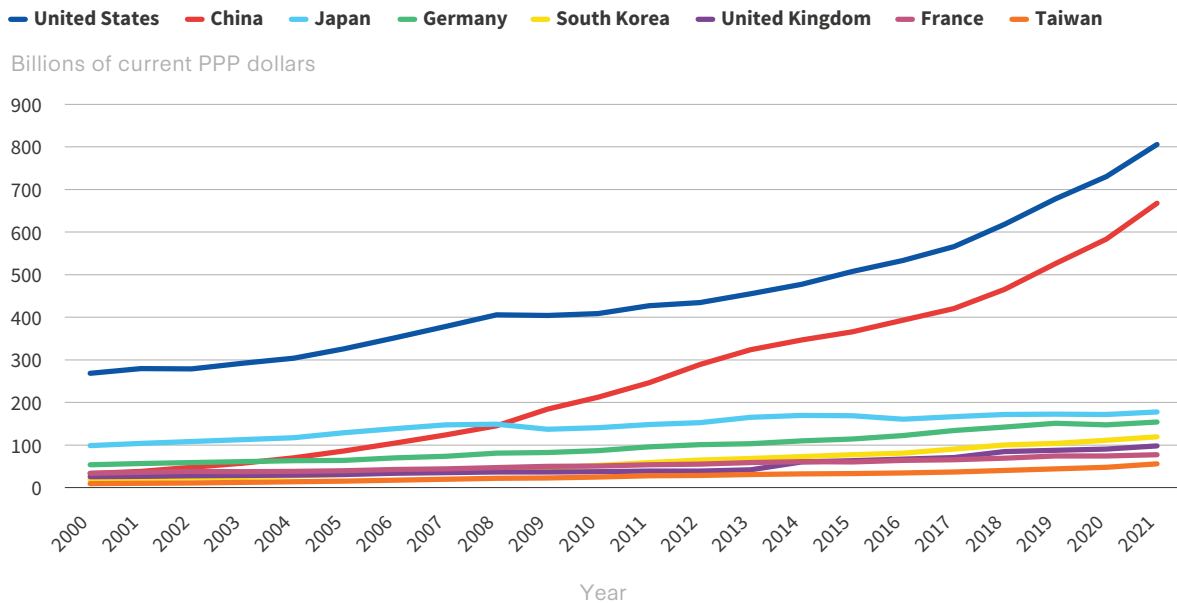
During the past two decades, wavering and erratic U.S. public support for science and technology stand in stark contrast with the massive and sustained Chinese effort to boost its innovation capabilities. In the 2010s, funding for R&D in the United States fell in real terms as federal spending was reduced across most categories. While these cuts satisfied short-term domestic political concerns, losses in R&D funding are not easily made up, and the effects endure. Emerging indicators suggest that China is closing the innovation gap with the United States, a development that should command the attention of U.S. policymakers and the public.

In March 2024, the NSF released a report showing that while the United States still leads the world in R&D spending, with \$806 billion in gross expenditures in 2021, China was close behind, at \$668 billion.¹³ As recently as 2009, U.S. R&D outlays were more than double those of China (Figure 1).

The NSF noted that with respect to indicators of global science, technology, and innovation (STI) capabilities, such as science and engineering (S&E) research publications, patents, and knowledge and technology intensive (KTI) industry output, “China has significantly increased its share of global STI capabilities.”¹⁴

China is now the top producer of S&E publications and international patents and has the largest value-added KTI manufacturing output, at \$2.4 trillion in 2021, versus the United States’ \$1.4 trillion.¹⁵

Figure 1: Gross Domestic Expenditures on R&D by Selected Country or Economy, 2000–2021



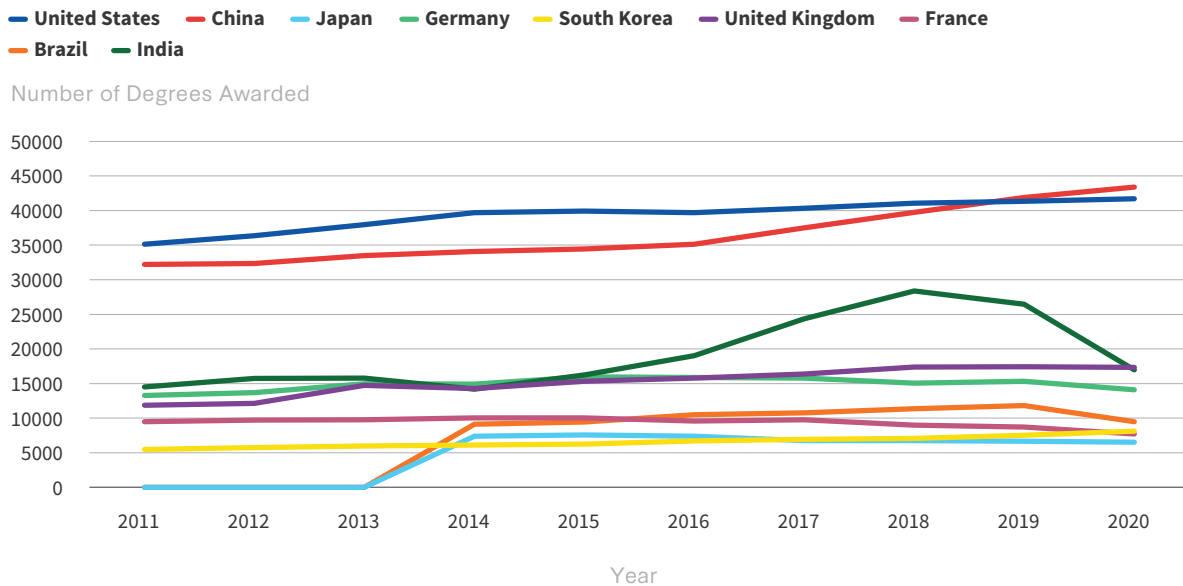
Source: Steven Deitz and Christina Freyman, *The State of U.S. Science and Engineering 2024* (Alexandria, VA: National Science Foundation, March 13, 2024), <https://ncses.nsf.gov/pubs/nsb20243/discovery-u-s-and-global-r-d#global-r-d>.

The United States has long led the world in doctorate degrees awarded each year in the S&E field (Figure 2). However, in 2020 China, with nearly 43,400 degrees, surpassed the United States, with around 41,700. While China’s larger population clearly impacts comparisons, China nonetheless far surpasses the United States in first university degrees—which are essentially equivalent U.S. bachelor’s degrees—in S&E fields, graduating two million in 2020, more than double the 900,000 in the United States.¹⁶ In the case of degrees, patents, and publications, qualitative differences may be significant, but the scale of China’s aggregated indicators are worth noting.

The Australian Strategic Policy Institute (ASPI) national security think tank, in a 2023 analysis of research trends in 23 strategic knowledge-intensive sectors related to Australian national defense, found that China led advanced technological research in 80 percent of the sectors. ASPI found that state-led investment allowed China to surpass the United States, Europe, and Japan in areas including hypersonic, sonar, advanced underwater communications, post-quantum cryptography, and underwater drones.¹⁷

A January 2023 study by the Information Technology and Innovation Foundation (ITIF), a U.S.-based think tank, found that “based on key indicators of innovation and advanced-industry performance, China has surpassed the United States in total innovation output.”¹⁸ Among the study’s key findings were that China is threatening U.S. and allied market share in advanced, high-value-added, and national security-critical industries. China is evolving “from an imitator to an innovator,” it said, and has demonstrated the capacity for world leadership in several advanced technologies, including supercomputers and high-speed rail.

Figure 2: S&E Doctoral Degrees Awarded by Selected Country, 2011–2020

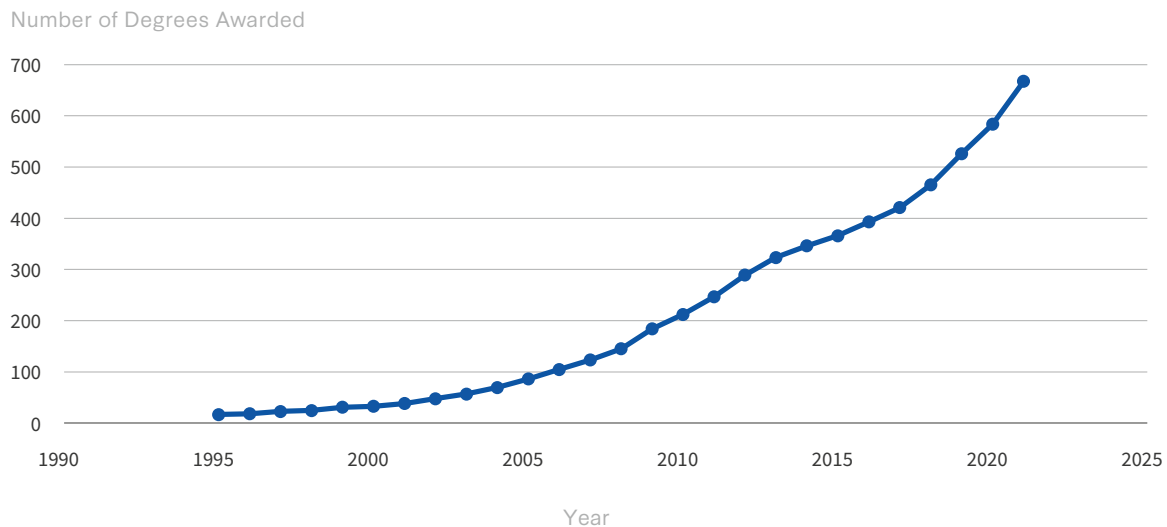


Note: Data for Brazil and Japan begins in 2014 due to lack of data for 2011–2013.

Source: Deitz and Freyman, *The State of U.S. Science and Engineering 2024*.

A study on Chinese innovation published by Cambridge University Press in 2023 characterized China as an emerging “juggernaut in science, technology and innovation,” citing a number of factors. According to the study, by 2021 China was spending double its 2015 R&D expenditure and 56 times the amount it spent in 1995, the year the government proposed a development strategy of “rejuvenating the nation with science, technology and education” (Figure 3).¹⁹

Figure 3: China Gross Domestic R&D, 1995–2021



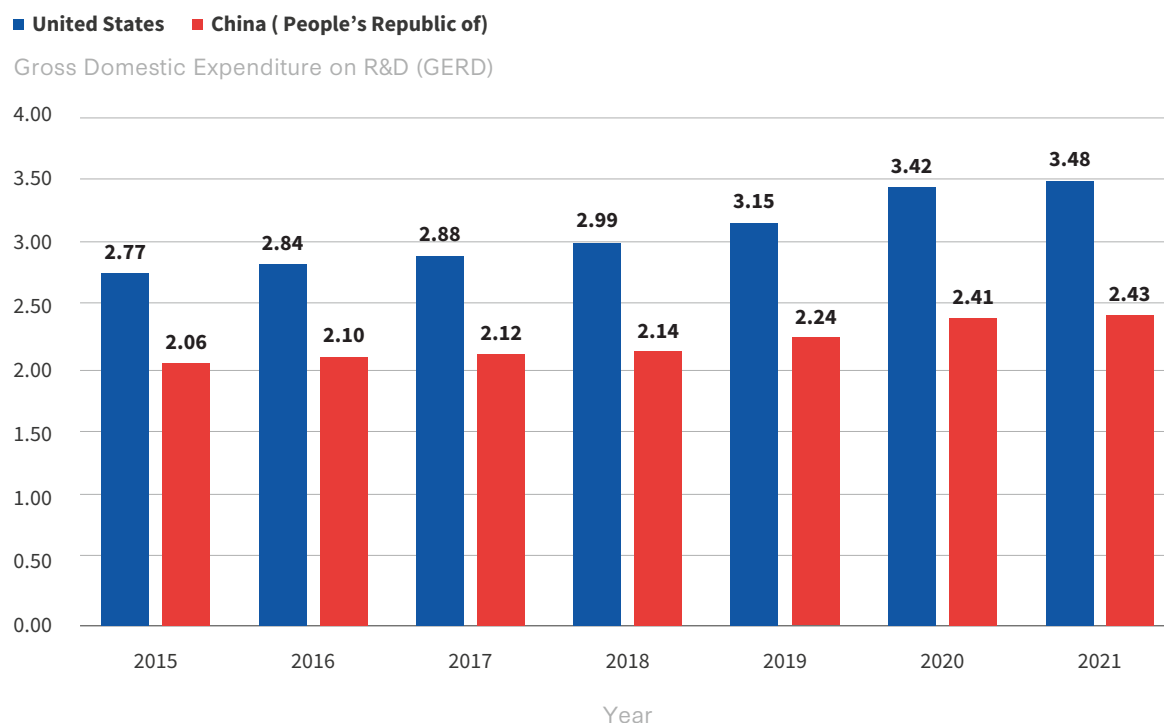
Source: Deitz and Freyman, *The State of U.S. Science and Engineering 2024*.

The study also found that China has the largest talent pool in the world: in 2020, the number of total undergraduate degrees awarded in China was 7.97 million, with graduate degrees totaling some 662,450 and doctorates 66,175.²⁰

A 2022 study of China's innovation policies published by Oxford University Press observed that “in absolute values, such as R&D expenditure and the number of researchers, patents, and publications, China is now ranked first or second in the world.” The study added, “Examples of its scientific and technological prowess include the development of the world's fastest supercomputer, China's first single-aisle jet aircraft, high speed rail networks, fifth-generation (5G) communications networks, and a demonstrated ability to land a spaceship on Mars, along with a plethora of world-leading innovations in the digital economy.”²¹

As mentioned previously, the United States continues to outspend China in gross domestic expenditure on R&D (GERD), although by a narrowing margin.²² The United States also outspends China in GERD as a percentage of GDP, spending 3.5 percent in 2021 compared to China's 2.4 percent.²³ However, these differences should not be grounds for complacency. It is notable that China's R&D allocation differs from that of the United States, with a greater proportion of spending focused on applied research and development and more limited efforts on basic research which, through legitimate and illegitimate means, can be acquired from U.S. sources (Figure 4).

Figure 4: Gross Domestic Expenditure on R&D as a Percentage of GDP, United States and China, 2015–21



Source: “Main Science and Technology Indicators,” OECD Data Explorer, accessed May 18, 2024, <https://data-viewer.oecd.org?chartId=80feaac0-9eac-4307-a324-921e8e1b9f70>.

Importantly, U.S. R&D spending is spread out over a vast array of commercially oriented themes and industries, while Chinese investments are concentrated in several dozen strategically important high-tech sectors. Most of these sectors have defense-related implications and are focused not on the pursuit of breakthrough innovations but on the dominance of those sectors through more efficient manufacturing and commercialization of recent innovations, most commonly drawn from R&D initially conducted outside of China.

China

An Unprecedented Degree of Government Engagement

While government policy measures have played a key role in the development of high-tech industries in the United States, Europe, and East Asia, “the Chinese case offers a rare and unprecedented opportunity to examine state-led innovation as few other countries have intervened so systematically and invasively in their national innovation systems,” the Cambridge University Press study noted.²⁴ In the United States, perennial internecine battles over the federal budget have led to continuing uncertainty over the scale and scope of the government’s innovation policies. In contrast, China’s government has pursued a consistent, long-range technology development effort for nearly two decades, albeit with adaptive adjustments for shocks such as the 2008 financial crisis, the Covid-19 pandemic, and the imposition of stringent technology export controls by the United States and its allies in and after 2022.²⁵

The Medium and Long-Term Plan

In June 2006, the Chinese government promulgated its “Medium and Long-Term Plan (MLP) for the Development of Science and Technology (2006-2020),” which the Office of the U.S. Trade Representative described in 2018 as “the seminal document articulating China’s long-term technology development strategy.”²⁶ The MLP featured a panoply of government measures to promote the development of “key fields and priority subjects,” including fiscal and tax measures, preferential government procurement, and “absorption” of imported technologies.²⁷

The MLP introduced the concept of “indigenous innovation,” defined as “enhancing original innovation through co-innovation and re-innovation based on the assimilation of imported

technologies.”²⁸ The concept of indigenous innovation has been carried forward in subsequent Chinese industrial developmental measures to the present day. However, indigenous is perhaps a misnomer. The 2018 study by the Office of the U.S. Trade Representative, which looked at China’s misappropriation of U.S. technology, offered extensive evidence of that assessment.²⁹

Sixteen “mega projects” were created in 2006 pursuant to the MLP to “help China master the core technologies of various strategic industries, driving the indigenous innovation effort.”³⁰ According to one source, the government planned to invest \$100 billion in these projects between 2006 and 2020.³¹ The megaprojects were tasked with the “assimilation and absorption” of technologies imported from outside China.³²

The MLP is a well-focused, well-funded, long-term effort supported by the Chinese government’s 11th, 12th, and 13th Five-Year Plans (2006-2020) for economic development. In addition, between 2006 and 2008, central government agencies published 79 innovation policies designed to implement the MLP, including sector-specific plans for key industries like semiconductors and renewable energy equipment.³³

Made in China 2025

Reflecting the continuity of these plans, in May 2015 the Chinese government announced its Made in China 2025 strategy, a 10-year, \$300 billion plan for promoting manufacturing that “brought together all existing policies across related fields, some of the ‘strategic emerging industries,’ and targeted priorities in automation, IT, robotics, AI, and much more under one umbrella.”³⁴ The Made in China 2025 strategy remains “consistent with the goal of ‘indigenous innovation.’”³⁵

It is also a remarkably ambitious strategy. With a goal of achieving global dominance in key technologies, it sets market share targets for designated industries and seeks to localize supply chains for those industries. China has set a target of being able to provide 70 percent of the key materials and components from domestic sources by 2025.

Funding for the strategy is equally ambitious. A prodigious volume of central and regional government funds flowed through many channels to participating firms in the form of equity investments, loans from government financial institutions, and huge public/private investment funds. Reflecting the government’s long-term commitment, China also established R&D centers across the country to develop priority technologies and support their commercialization.

In addition, China has launched new large-scale “mega” or “high end” thematic R&D projects—described as “large scale infrastructures and expensive funding initiatives in strategic industries and frontier technologies”—continuing a policy tool introduced under the 2006 MLP.³⁶

China has also set targets for achieving “green” and “smart” manufacturing in the key sectors, incorporating digital technologies in the effort, with the government designating “pilot demonstration projects” across the country to facilitate dissemination of newly developed technologies.³⁷ These indigenous investments are complemented by government-supported

investment funds that acquired foreign firms possessing technologies deemed to be supportive of China's development effort.³⁸

Significant Economic Impact

China's actions have proven effective and have even generated notable successes. As Professor Shirley Ze Yu, a political economist at Harvard's Kennedy School, observed in March 2024, "Made in China 2025 is an exceptionally successful industrial strategy."³⁹ She went on to note, "In six years, China's smartphone manufacturing surpassed 50 percent of the global market share. In eight years, China's solar, wind and lithium battery production all rose to global dominance." Although the Made in China 2025 strategy was initially "grossly underestimated," it is now "considered in the West, especially in the United States, as a master plan for China to secure dominance in global technology, connected with the expansion of its global influence using the Belt and Road Initiative."⁴⁰

China's Innovation Vulnerabilities

Despite this progress, the widespread view that China's bid for global technological leadership is relentless and unstoppable has its share of skeptics, and in fact the recent record of innovation successes masks systemic weaknesses.

Reliance on Foreign Research

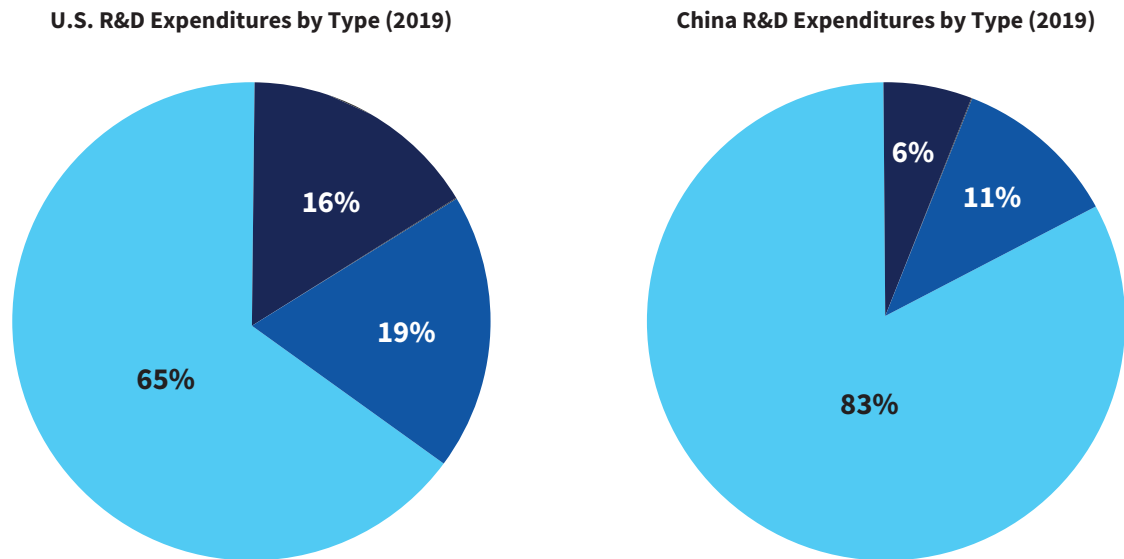
China's total R&D spending, based on 2019 figures, allocates only around 6 percent of total outlays to basic research and 11 percent to applied research, compared with 16.6 percent and 19.2 percent, respectively, for the United States (Figure 5).⁴¹

The remainder is allocated to “experimental development”—that is, according to the NSF's definition, “drawing on knowledge gained from research and practical experience and producing additional knowledge, which is directed to producing new products or processes or to improving existing products or processes.”⁴² Underfunding of basic and applied R&D presumes the continued ability to absorb foreign technology and manufacture products based on it. China's R&D overspending on development at the expense of basic and applied research “may jeopardize the long-term prospect of the country's scientific, economic, and social development,” scholars have found.⁴³

To date, the indigenous innovation approach, drawing upon the fruits of foreign research, may have enabled China to escape the consequences of its underdeveloped basic and applied research efforts. As the Mercator Institute for China Studies (MERICS), a European think tank specializing in China,

Figure 5: R&D Expenditure by Type, United States and China, 2019

■ Basic ■ Applied ■ Experimental development



Source: Deitz and Freyman, *The State of U.S. Science and Engineering 2024*.

observed in March 2024, “Most of the [technology] successes China can point to have some degree of foreign inspiration. Digital giants in search, e-commerce, ride hailing and social networking started by translating U.S. models to Chinese contexts.” MERICS went on to point out, “In hardware sectors like solar panels, batteries, electric vehicles, smart phones, and commercial drones, Chinese overseas returnees and local entrepreneurs gained global market share by building on ideas and components pioneered elsewhere.”⁴⁴

But as recent events have underscored, China’s degree of access to foreign technology, especially at the cutting edge, will not necessarily continue. The Biden administration is reportedly considering blacklisting Chinese semiconductor firms linked to Huawei Technologies and imposing sanctions on Chinese memory chipmaker ChangXin Memory Technologies.⁴⁵

In particular, “access to core components and technology is a prerequisite for China’s advancement in emerging industries,” MERICS scholars wrote in 2019. “Chinese tech firms have already gotten into serious trouble when cut off from access to chips or other high-tech components from abroad, as U.S. measures towards companies like ZTE and Huawei have proved.”⁴⁶ The recent imposition of more stringent export controls on semiconductors, quantum computing, and AI technology by the United States and its allies has underscored this vulnerability.⁴⁷

Encumbrances of Government Involvement

While the Chinese government’s deep involvement in innovation is often cited as a source of strength, it is also problematic. The 2023 ITIF study, which warns of China’s growing prowess in this regard, acknowledges that “state involvement in the Chinese economy will almost certainly hinder

productivity growth.”⁴⁸ But it is important to keep in mind that China is not seeking to maximize efficiency; rather, it strives to dominate strategic sectors with national security implications, and it is willing to spend heavily to do so.

Still, Chinese scientists sometimes complain that as they seek funding, they are “encumbered by arduous red tape,” while their peers are “becoming obsessed with recognition,” undermining the nation’s drive for innovation and technological self-sufficiency.⁴⁹

For example, in August 2023, China’s Ministry of Industry and Information Technology decreed that all mobile app providers must submit their business details to the government or face punishment, a move that Chinese software developers said will “stifle local innovation and hinder access to advances created overseas.”⁵⁰

Limited Freedom to Think and Associate

While China is graduating more students in S&E fields, it remains uncertain whether sheer numbers will be sufficient to move China ahead in innovation. One recent analysis observed that

while innovation has been elevated to a very high status in China, and on the surface Chinese researchers are encouraged to think outside the box, equally important are the other ingredients of a true innovation culture, namely, autonomy, free access to the flow of information and especially dissent, both scientific and political, which at present are not adequately applauded or tolerated.⁵¹

Growing Friction with Trading Partners

Huang Yiping, dean of Peking University’s National School of Development, warned recently that China’s massive investments in high-tech manufacturing sectors could lead to overcapacity and friction with China’s trading partners, noting that the United States and Europe were moving to decouple their supply chains from China and invest in increased domestic production.⁵² “If it really turns into a relatively common wave of trade protectionism against Chinese products, it may actually be detrimental to our next stage of development, especially in innovation,” he said.

Yet while it is true that China’s innovation system has weaknesses and vulnerabilities, China can point to spectacular achievements during the past decade in numerous technology-intensive fields. In virtually every case, China got its foothold through acquisition and adaptation of foundational foreign technology. But it has gone on to develop and improve manufacturing capability, frequently surpassing its foreign counterparts in quality and efficiency and eventually achieving a dominant global market position. As a 2023 study pointed out, “While the development of novel products and processes is obviously an important aspect of innovation, effective commercialization is at least as important.”⁵³ And successful commercialization is essential for the revenue to fund the R&D to drive the next wave of innovation.

A Growing Challenge in Strategic Sectors

In 2023, CSIS published a major study of the new controls imposed by the United States and its allies on technology exports to China, whose coauthors included William A. Reinsch, formerly in charge of U.S. export controls at the Department of Commerce. Although the study's focus was on export controls, it concluded by stating that “the important question is not how to hold China back [through controls], as the U.S. capability for doing that is limited, but rather how to stay ahead. A strategy for that lies in the various U.S. policy initiatives enacted in the past two years, most notably the CHIPS Act, but that alone will be insufficient to achieve long-term U.S. strategic objectives.”⁵⁴

At present, the policies adopted amid much fanfare to address this existential competition are effectively stalled. The “science” portion of the CHIPS Act—the principal longer-term scientific promotion measure of the current administration—remains unfunded, a circumstance that may well continue for the foreseeable future. Congress, instead of dramatically expanding funding for federal science agencies, as in the Sputnik era, is reducing it. This is not an abstract problem. A survey of three critical cross-cutting foundational technologies that will enable many strategic technology sectors essential for U.S. national defense capabilities—AI, quantum technology, and semiconductors—indicates that while the United States may still lead in all three, China is rapidly closing the technology gap.

Artificial Intelligence

AI is the development of computer systems to perform tasks and exercise cognitive functions that usually require human intelligence, such as decisionmaking, pattern recognition, problem solving, “learning” from available data, and translation of languages. It is rapidly being deployed across a broad range of human endeavors, transforming economies and societies.⁵⁵ A high-stakes AI race, reminiscent of the U.S.-Soviet race to the moon in the 1960s, is emerging between the United States and China. Unlike the race to the moon, however, the race for AI has no short-term finish line. On the contrary, it involves competition across national innovation ecosystems including universities, small and large companies, and government funding and policy support.

China recognizes this is a long-term contest and is taking a long-term view, with investments commensurate with the stakes. For example, in 2017 China’s State Council released its New Generation Artificial Intelligence Plan, which set 2030 as a target date by which China will achieve global leadership in AI, turn AI into a \$150 billion domestic industry, and be the world leader in developing ethical norms and standards for AI.⁵⁶

WHAT ARE THE STAKES IN THE AI RACE?

Policy experts increasingly recognize the importance of AI in the context of national defense.⁵⁷ AI can make multiple decisions and enable actions with a speed and performance level that, in the future, is expected to exceed the capabilities of human operators. “As AI does not get fatigued, does not forget, and has no emotional fluctuation, AI is expected to be able to help commanders make decisions by processing large quantities of data quickly and accurately,” the Japanese National Institute of Defense Studies said in 2021.⁵⁸

The U.S. National Security Commission for Artificial Intelligence (NSCAI) was formed in 2018 to assess AI, machine learning (ML), and related technologies as they may affect U.S. national security and defense, and to make recommendations to the president and Congress. The commission released its final report in March 2021, which concluded that the United States “is not prepared to defend or compete in the AI era. . . . This is the tough reality that we must face,” in the words of Chair Eric Schmidt, former chief executive of Google, and Vice Chair Bob Work, a former deputy secretary of defense.⁵⁹

The NSCAI warned that “for the first time since World War II, America’s technological predominance—the backbone of its economic and military power—is under threat. China possesses the might, talent, and ambition to surpass the United States as the world’s leader in AI in the next decade if current trends do not change.”⁶⁰ The commission said that “our armed forces’ competitive military-technical advantage could be lost within the next decade if they do not accelerate the adoption of AI across their missions.”⁶¹ It also emphasized the potential consequences, asserting that “defending against AI-capable adversaries operating at machine speeds without employing AI is an invitation to disaster.”⁶²

The NSCAI noted that while the U.S. private sector was making dramatic strides in AI, “commercial agendas are dictating the future of AI, and concentrating heavily in one discipline: machine learning (ML).” It added, “Despite promising moves, government funding has lagged behind the

performative potential of the field, limiting its ability to shape research toward the public good and support progress across a range of AI disciplines. As a result, the [U.S.] AI innovation environment rests on a narrowing foundation.”⁶³

Although the NSCAI’s report was released three years ago, recent comments by industry and government experts indicate that the commission’s basic findings with respect to the national security dimensions of AI remain sound. This is reflected in both the scale of Chinese investments and their focus, which prioritize AI-related research in defense applications instead of consumer services such as the ChatGPT chatbot: Alexandr Wang, the founder of Scale AI, said in 2023 that while China’s People’s Liberation Army is spending between 1 and 2 percent of its annual budget on AI, the Pentagon is spending between 0.1 and 0.2 percent.⁶⁴

In mid-2023, Work, the former deputy defense secretary, commented that in “marginal terms, we’re flat and the Chinese are outspending us. This is why China is so different. In the past we’ve always been able to outspend our competitors.”⁶⁵

Deputy Assistant Secretary of Defense Michael Horowitz said in a January 2024 interview that “the [AI] adoption capacity of the Department is improving . . . but we have more work to do, frankly, as we’ve been very public in stating.”⁶⁶

PAYOFFS FOR CHINA

China is making great strides in developing defense-related applications of AI. The 2023 ASPI assessment of research trends in 23 critical technology areas concluded that in six key thematic areas relevant to AI and autonomy, China already leads in three (Table 1).⁶⁷

Table 1: Research Leaders in Defense-Related AI Applications

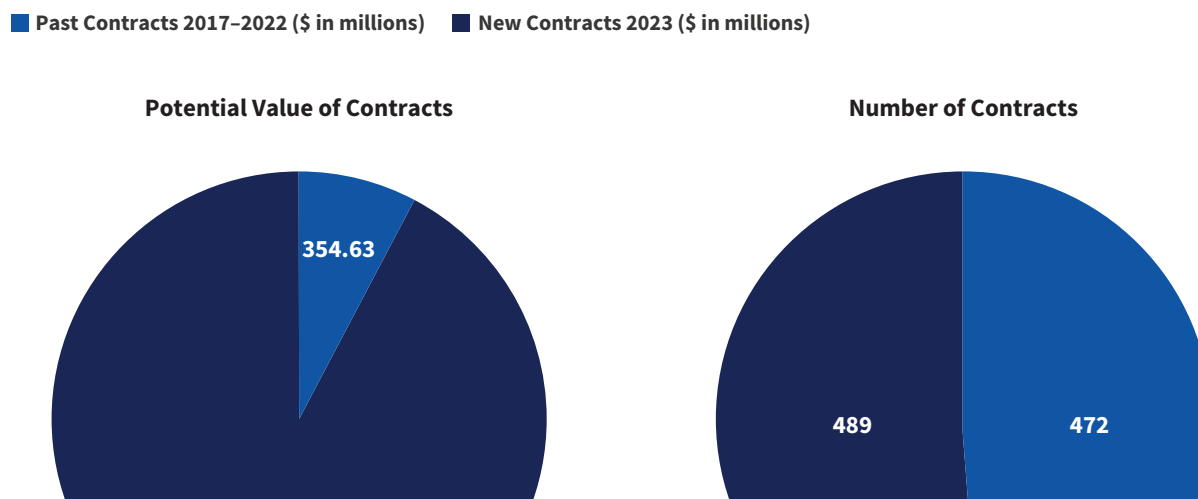
AI Theme	Leading Country	Top Research Organization
Drone swarming and collaborative robots	China	Beihang University
AI algorithms and hardware	China	Tsinghua University
Advanced data analytics	China	Chinese Academy of Sciences
Machine learning systems	United States	University of California
Advanced integrated circuit design/fab systems	United States	University of California
Natural language processing	United States	Google (U.S. Division)

Source: “AUKUS Relevant Technologies: Top 10 Country Snapshot,” Australian Strategic Policy Institute, Critical Technology Tracker, June 2023, <https://www.aspi.org.au/report/critical-technology-tracker>.

The Department of Defense (DOD) is beginning to respond to the AI challenge. In 2022, Brookings released a study of all federal AI-related contracts in the preceding five years, and the next year it released a follow-up study covering August 2022 to August 2023.

Over a five-year period from 2017 to 2022, it found a total of 472 AI-related federal contracts. In just the one-year span from 2022 to 2023, there were 489. The potential value of federal awards for AI research grew 1,200 percent in the 2022-23 period, to \$4.6 billion, from \$355 million over the preceding five-year period, primarily driven by the DOD (Figure 6).⁶⁸

Figure 6: Comparing Potential Value and Contract Counts, Past vs. New Contracts



Source: Jacob Larson et al., *The Evolution of Artificial Intelligence (AI) Spending by the U.S. Government* (Washington, DC: Brookings, March 26, 2024), <https://www.brookings.edu/articles/the-evolution-of-artificial-intelligence-ai-spending-by-the-u-s-government/>.

However, as reported in *Time*, the Pentagon questioned the Brookings analysis, stating that DOD requested \$874 million for AI research, development, testing, and evaluation (RDT&E) in FY 2022 and \$1.8 billion in FY 2024.⁶⁹

In either case, the U.S. defense establishment is clearly increasing its commitment to AI, although whether this will match China’s effort is unclear. In its November 2023 strategy to promote the adoption of AI by the armed forces, the Pentagon’s stated goal was to “help accelerate the speed of commanders’ decisions and improve the quality and accuracy of those decisions, which can be decisive in deterring a fight and in winning a fight.”⁷⁰

Deputy Secretary of Defense Horowitz disclosed in his January 2024 interview that with a program dubbed the Replicator Initiative, the Pentagon expects to be able to “field in the multiple thousands attritable autonomous systems in the next 18 to 24 months. . . . We’re on track to achieve that goal.”⁷¹ This could mean the deployment of thousands or tens of thousands of low-cost autonomous units for use in swarms which, from a cost standpoint, the United States could “afford to lose.”

Matt Turek, deputy director of the Information Innovation Directorate at the DOD’s Defense Advanced Research Projects Agency (DARPA), said at a March 2024 CSIS event that about 70 percent of all DARPA initiatives now have some form of AI, machine learning, and autonomy associated with them, with the broad goal of “preventing or creating strategic surprise.”⁷² He noted that technology

firms like Microsoft, OpenAI, Anthropic, and Google were all participating in a DARPA program and were providing access to state-of-the-art models.

Despite such developments, as former deputy secretary of defense Work said in mid-2023, while the Pentagon is increasing its R&D spending on AI, it is not doing so at a pace sufficient to close the gap with China.⁷³

CREATING AN INSTITUTIONAL ARCHITECTURE

The Biden administration is creating an institutional architecture for the application of AI throughout the federal system. In October 2023, the administration released an executive order on AI that emphasizes safety, governance, and the establishment of standards and consumer protections; sets up task forces and advisory committees; and requires specific actions by big tech companies and federal agencies, including the promulgation of guidelines.⁷⁴ The administration is requesting \$3 billion for federal agencies' use of AI in its FY 2025 budget and \$300 million to increase agency funds for AI to address risks and advance AI's use for the public good.⁷⁵

The budget allocates \$32 million to a National AI Research Resource (NAIRR) pilot, which will provide AI researchers with the computational tools they need to conduct AI research.⁷⁶ Federal agencies would get \$70 million to establish chief AI officers (CAIO) to promote the use of AI and manage its risks. Meanwhile, an "AI talent surge," aimed at boosting the federal talent pool with AI expertise, is allocated \$32 million.⁷⁷

While these measures will undoubtedly facilitate the application of AI in the federal agencies, they appear to be primarily focused on safety measures, consumer protection, and safeguards against abuses of AI. They are not focused on the strategic AI competition with China.

AI CHALLENGES FOR CHINA

Despite its gains in AI, China faces obstacles in its effort to capture world leadership in the field by 2030. While the government has been a critical actor in promoting the development of AI, its authoritarian character is, at times, counterproductive.

In the summer of 2023, Chinese authorities proposed regulations for generative AI, requiring that all images and text must align with the "core values of socialism" and must not undermine state authority or national unity.⁷⁸ At that time several Chinese companies had already launched chatbots but found it impossible to compete with U.S.-based OpenAI's ChatGPT in part because of the many restrictions imposed by the state.

A major U.S. advantage is that the private sector is investing heavily. According to the Stanford University Institute for Human-Centered AI's 2024 annual report, the United States leads China in private sector investment in AI by an enormous margin, with \$67.2 billion invested in 2023 compared with \$7.8 billion by China.⁷⁹ U.S.-based "big data" companies like Meta, Google, and Microsoft are pouring large sums of money into AI research. The United States is also spawning AI start-ups at over three times the rate of China. OpenAI, a U.S. nonprofit AI research organization, launched ChatGPT in 2022 powered by a large language model, and it became the fastest-growing consumer software application in history.⁸⁰

Recently imposed Western export controls are limiting China's access to the leading-edge chips that are essential to drive advanced AI systems. China is unlikely to be able to produce such chips in volume by itself for a number of years, although Huawei's recent announcement of seven-nanometer technology in its latest smartphone is a troubling sign for U.S. policymakers.⁸¹ Still, Paul Scharre, director of studies at the Center for a New American Security (CNAS), noted that if the Western controls work as intended, "Chinese AI developers will remain boxed out of that supply chain, working with less powerful chips, and thus a year or two behind the cutting edge as AI continues its rapid advance."⁸²

These controls can be effective in delaying China's progress in AI. According to numerous industry experts, "Even as the country [China] races to build generative A.I., Chinese companies are relying almost entirely on underlying systems from the United States," and China lags behind the United States by a year and may be falling further behind in generative AI.⁸³

China faces obstacles in developing large language models, such as the relatively lower quality of data from Mandarin-language internet sources versus English and strong competition from Google's Gemini and ChatGPT, but is apparently finding workarounds.⁸⁴ In March 2024, the Department of Justice indicted a Chinese national and former Google AI software developer for allegedly stealing 500 files of confidential code that the company uses for its supercomputing data centers to train large language models.⁸⁵ The indictment alleged that the researcher concurrently went to work for Google rivals in China. Indeed, a hidden Chinese competitive advantage in AI and other high-tech sectors is its large-scale state-supported theft of Western intellectual property.

Quantum Information Science

Quantum information science (QIS) employs principles derived from quantum science to achieve new and potentially revolutionary capabilities in communications, computing, sensing, and encryption/decryption. Quantum information technology has rapidly moved from pioneering theoretical work by U.S. scientists to the emergence and pursuit of practical applications in the United States, China, and Europe. Many of these applications will have major impacts on national security.

For instance, quantum computers may soon be able to decrypt information stored on encrypted systems, which, according to a 2022 Biden administration National Security Memorandum, could "jeopardize civilian and military communications systems, undermine supervisory and control systems for critical infrastructure, and defeat security protocols for most Internet-based financial transactions."⁸⁶

Among other uses, application of quantum concepts to sensors could enable the detection of concealed underground structures, submarines, and nuclear weapons.

Quantum-based communications systems could ensure secure communications that could not be breached by adversaries. And quantum accelerometers could be used for navigation when GPS is not available, including underwater and underground situations and scenarios in which GPS has been knocked out by adversaries.

THE U.S. NATIONAL QUANTUM INITIATIVE ACT

In a major positive step to address this challenge, the National Quantum Initiative Act (NQIA) of 2018 established a coordinated federal program to promote quantum R&D to ensure U.S. economic and national security.⁸⁷ The act authorized federal agencies to establish consortia and research centers to enable QIS R&D and required the coordination of QIS programs across the federal government and in conjunction with academia and industry.⁸⁸ The NQIA was authorized for five years and is up for reauthorization in 2024.

Reauthorization of the act is crucial. As a principal analyst at UK-based data analytics firm GlobalData observed, “If reauthorization fails, it will damage the U.S.’s position in the global quantum race.”⁸⁹

Significant QIS developmental efforts are ongoing at over a dozen federal agencies including NIST, DOD, DOE, NASA, and within the intelligence community.⁹⁰ DOD’s QIS developmental efforts have been under way for years and are implemented pursuant to the National Defense Authorization Act. Since 2023, DARPA has been pursuing Underexplored Systems for Utility-Scale Quantum Computing (US2QC) to determine whether revolutionary approaches to QIS can reach utility scale more rapidly than conventional wisdom predicts. There are multiple paths to pursue. These opportunities were recognized in the increased spending authorized in the science portion of the CHIPS Act for quantum research at NSF, NIST, and DOE, but as noted above, the necessary funds have not been appropriated.

A GROWING COMPETITIVE THREAT

Where the United States and China stand in comparative terms in QIS is subjective and difficult to assess, particularly given that the research efforts that are under way are frequently not transparent.

A 2022 RAND Corporation study concluded that the United States led the world in most, but not all, quantum technologies.⁹¹ In 2022, GlobalData concluded that the United States led China in quantum technology by about five years, but two years later, in 2024, the same firm concluded that the two countries were “nearly equal.”⁹² Underscoring this assessment, in May 2024, research teams in China, the United States, and the Netherlands independently reported simultaneous breakthroughs that could eventually enable the creation of “nearly unhackable” quantum-based internet services.⁹³ The 2023 Australian ASPI survey of critical technologies found that of four quantum-related research domains, China led in three: quantum sensors, quantum communications, and post-quantum technologies.⁹⁴ The United States leads in one defense-related field, quantum computing. In the three fields led by China, that country is producing more S&E papers and has the foremost research organizations in the world, the Chinese Academy of Sciences and the University of Science and Technology of China.

According to a 2023 study by the Wilson Center, “opacity reigns in China’s public and private financing for quantum technologies.”⁹⁵ The same can be said, to a degree, about U.S. outlays.

The Chinese government has committed to spend \$15 billion in developing QIS between 2024 and 2028, but the extent to which it will actually do so has been questioned.⁹⁶ In the United States, the

government has reportedly committed \$3 billion to various quantum projects, another \$1.2 billion to the National Quantum Initiative, and possibly additional outlays in the DOD's mostly classified "black budget."⁹⁷ Although U.S. public outlays are smaller than China's, as a recent CSIS study observes, "The United States has begun to mitigate its investment gap with China by teaming up with key allies leading global QIS innovation on research and development efforts."⁹⁸

MAJOR ACHIEVEMENTS, YET VULNERABILITIES REMAIN

Recent anecdotal reports out of China underscore both its achievements in quantum technology and some of its vulnerabilities. In October 2023, Chinese scientists announced that the country's latest quantum computer, the JiuZhang 3, had solved a highly complex mathematical problem in a millionth of a second, more than 20 billion years faster than the world's fastest supercomputer could solve the same problem.⁹⁹ In January 2024, a Chinese quantum computer company, Origin Quantum—founded by scientists from the University of Science and Technology of China—unveiled a "China independent-developed state-of-the-art quantum computer" dubbed Origin Wukong (Monkey King) which was open for access by users around the world.¹⁰⁰ Within 10 days, the computer had performed 33,871 quantum computing tasks for global users. "No other nation funds more R&D for [quantum] communications than China," the Wilson Center study noted.¹⁰¹

Chinese research institutes are reportedly creating a quantum communications network using satellites in low and medium-to-high Earth orbits.¹⁰² The network would utilize quantum technology for encryption and secure transmission of data and communications.

In March 2024, Guo Guoping—a quantum physicist, cofounder of Origin Quantum, and leading figure in China's quantum promotional effort—warned that China was dependent on Western countries, mainly the United States, the Netherlands, and Finland, for dilution refrigerators, which are used to create the ultra-low temperatures required by quantum computing.¹⁰³ Guo also noted that China lagged behind the West in industrial applications of quantum computing, citing high costs and inadequate market demand.

Semiconductors

All major U.S. defense platforms, domestic infrastructures, and technology-intensive products and industries are based on semiconductor chips, the basic building blocks of digital technology.¹⁰⁴ AI systems, for example, "operate on a foundation of interconnected computer hardware driven by cutting-edge semiconductor devices." The U.S. semiconductor design industry leads the world, representing a crucial U.S. asset in the strategic competition with China.

The CHIPS Act of 2022 was intended to address twin U.S. vulnerabilities in the chip manufacturing sector that were revealed by the Covid-19 pandemic—most notably the shortfall of U.S. production capacity needed to supply the needs of domestic industry and the U.S. military, and the fact that U.S. chipmakers trail Taiwan and South Korea in their ability to make chips at the most advanced nodes. At present, the United States is dependent upon chipmaking facilities in Taiwan, which makes 90 percent of the most advanced chips, and South Korea for these devices. Taiwan, of

course, faces significant levels of geopolitical risk but also seismological ones: the island suffered several earthquakes in April 2024 alone.¹⁰⁵

After a necessary ramp up, CHIPS Act funds are now being awarded to firms to expand both leading-edge and foundational semiconductor manufacturing capacity in the United States. Moreover, the U.S. government, in conjunction with key allies, is enforcing new and more stringent controls on exports of chips and chip manufacturing technology to China, especially on tools and devices necessary to support advances in AI.¹⁰⁶

China has responded by redoubling its commitment to developing its own semiconductor technology through domestic innovation. At the end of 2022, the Chinese government was reportedly planning to spend 1 trillion yuan (\$143 billion) over the next five years to support its semiconductor industry, more than the United States and the European Union combined.¹⁰⁷ In 2014, China established the China Integrated Circuit Industry Investment Fund (the “Big Fund”) to support investments in the domestic semiconductor industry, comprised of government, quasi-public, and private funds. By 2019, the Big Fund had raised 204 billion yuan (28 billion USD). In May 2024, China launched a new phase of this effort with a registered capital of \$47.5 billion to support investments in China’s semiconductor value chain, including production capacity, equipment, materials, and advanced packaging.¹⁰⁸

China is rapidly expanding its production capacity for higher node chips, sometimes called “foundational” semiconductors.¹⁰⁹ These chips account for most current applications worldwide—and Chinese firms may come to dominate that market segment to the detriment of U.S.-based chipmakers, including those currently investing in new facilities with support from the CHIPS Act.

For the moment, the U.S.-based semiconductor design industry leads all other countries, including China, by virtually any measure of international competitiveness. Advanced artificial intelligence chips designed by the U.S. fabless firm Nvidia account for over 80 percent of the market and “underpin all of the most advanced AI systems.”¹¹⁰ Nvidia reportedly holds a 92 percent global market share in data center graphics processing units (GPUs).¹¹¹ But in June 2024, a Huawei executive stated that the company’s Ascend 910B AI chip can deliver 80 percent of the efficiency of an Nvidia A100 when training large language models, and, “in some other tests, the Ascend can beat the A100 by 20 percent.”¹¹² The United States has a number of world-leading integrated device makers (IDMs) that design and manufacture their own chips, such as Intel Corporation and Texas Instruments. Intel is reportedly pursuing the astonishing goal of incorporating 1 trillion transistors on a single processor package by 2030, about 10 times the current number.¹¹³

In addition, U.S.-based firms dominate crucial upstream chip design sectors such as electronic design automation (EDA) tools and IP blocks, which enable faster and less expensive chip designs.¹¹⁴ Loss of access to these enabling technologies as a result of Western export controls is proving disruptive to China’s AI sector.

Some experts regard China’s prospects for achieving technological leadership in semiconductors through its own efforts as impossible, while others see it as inevitable. If a loss of leadership in

semiconductor design did occur, it would be a strategic calamity of monumental proportions for the United States, with a more far-reaching impact than the offshore movement of chipmaking that has been unfolding since the 1990s. Once lost, such leadership would be difficult to recapture. And as the 2021 NSCAI report warned, “If a potential adversary bests the United States in semiconductors over the long term or suddenly cuts off U.S. access to cutting-edge chips entirely, it could gain the upper hand in every domain of warfare.”¹¹⁵

For that reason, in addition to building out domestic chipmaking capacity, U.S. policymakers should be sensitive to the health and competitive standing relative to China of the domestic chip design industry, with an eye to preserving and even increasing U.S. leadership. The United States cannot be complacent about its current leadership. In that regard, in 2022 the Semiconductor Industry Association and the Boston Consulting Group warned that the U.S. chip design industry’s global market share has been eroding as design firms face challenges such as rising costs, labor shortages, and exclusion from important foreign markets.¹¹⁶

Conclusion

It is crucially important to understand that AI, QIS, and semiconductors are increasingly interrelated and mutually supporting fields. Over time, the advances in these domains may transform both national economic capabilities and the strategic balance. The interdependencies are powerful.

AI systems run on advanced chips, and AI software is increasingly being deployed to design those chips. Researchers are identifying composite materials that could integrate quantum devices into semiconductor technology, making chips “significantly more powerful.”¹¹⁷ Semiconductor technology is in turn being used to fabricate qubits, the basic building blocks of quantum computers.¹¹⁸ AI systems based on quantum processing could “be millions of times faster than the fastest microchip computers today,” experts say.¹¹⁹

Chips, AI, and QIS will therefore not only advance all twenty-first-century technology-intensive industries but also drive, enhance, and possibly disrupt each other. Chinese authorities understand this competition and are dedicating massive amounts of funding toward capturing the benefits of these transformative technologies.

Given the stakes of this competition, it is jarring that in the space of just a few days in March 2024, the United States both failed to fund the science portion of the CHIPS Act and implemented budget cuts for key federal science agencies.¹²⁰ Meanwhile, the Chinese government increased planned R&D spending by 10 percent. China has pledged to “harness the entire nation’s resources to speed homegrown breakthroughs, reaffirming what it sees as a central priority to become self-reliant

in spheres from AI to chipmaking and ultimately wrest technological supremacy from the West,” experts say.¹²¹

Sudip Parikh, chief executive of the nonprofit American Association for the Advancement of Science, warned that the race for leadership in key technologies like AI, quantum, and fusion would have long-term strategic implications: “Once you lose the lead in that kind of a race, the benefits accrue to whoever’s in the lead, and they’ve been accruing to us for the last 75 years. They will begin to accrue somewhere else.”¹²² U.S. policymakers need to recognize that danger and make the investments needed to secure the nation’s future.

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Sujai Shivakumar directs the Renewing American Innovation (RAI) at the Center for Strategic and International Studies (CSIS), where he also serves as a senior fellow. Dr. Shivakumar brings over two decades of experience in policy studies related to U.S. competitiveness and innovation. Previously, he directed the Innovation Policy Forum at the National Academies of Sciences, Engineering, and Medicine and led major studies of U.S. policies supporting advanced manufacturing, small business growth, workforce development, and entrepreneurship. He was also a lead contributor to a seminal National Academies study of strategies adopted by U.S. states and regions to foster entrepreneurship, drive technology transfer, and encourage regional high-tech ecosystems. He also helped prepare National Academy of Public Administration studies on laboratory technology transfer and the management of space situational awareness. Reflecting his expertise in innovation policy, Dr. Shivakumar has testified before the Senate Committee on Commerce, Science, and Transportation and has been quoted in leading publications such as the *Financial Times* and the *Wall Street Journal*. His academic background includes a doctorate in economics from George Mason University and service as an Earhart Foundation scholar at the Ostrom Workshop at Indiana University Bloomington, where he authored *The Constitution of Development* (Palgrave Macmillan, 2005) and coauthored with Nobel Laureate Elinor Ostrom *The Samaritan's Dilemma: The Political Economy of Development Aid* (Oxford University Press, 2005). During his tenure at the National Academies, he contributed to over 50 assessments of U.S. and foreign innovation programs.

Charles Wessner is currently a research professor at Georgetown University, where he teaches global innovation policy. He is active as a speaker, researcher, and writer with a global lens on innovation policy and frequently advises technology agencies, universities, and governments on

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Thomas Howell is an international trade attorney (currently in solo practice) serving as a consultant to the CSIS Renewing American Innovation. During the course of his 40-plus-year legal career, he has represented U.S.-based semiconductor companies and organizations in matters such as the U.S.-Japan trade disputes and litigation of the 1980s; the formation of Sematech in 1986-87; trade disputes with China (including the first WTO dispute settlement challenge to that country in 2003); and numerous other public policy initiatives. He served as a member of the Defense Science Board on Secure Microchip Supply in 2004. He has also represented U.S.-based industries in trade disputes and litigation in telecommunications equipment, soda ash, photographic film and paper, flat-rolled steel, and renewable energy equipment. In conjunction with his legal work, he developed a specialization in field research in Asia and Europe to produce analytic studies for clients. His subjects include industrial, research, science, and innovation policies outside the United States and foreign market access barriers to U.S. exports and investment. The National Academy of Sciences has commissioned and published a number of his works, and he has contributed to the academy as a contract writer and researcher producing numerous peer-reviewed chapters and book-length studies. He has published many articles and books, the most recent of which was coauthored with Charles Wessner, *Regional Renaissance: How New York's Capital Region Became a Nanotechnology Powerhouse* (Springer Nature, 2020). He is a graduate of Harvard College (1971) and Boston University School of Law (1977).

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